INDUSTRIAL EQUIPMENT INSPECTION REPORT

Synthetic Sample Document — For Sovereign AI Workbench Testing

Document ID	IR-CWP-2026-014	**Report Type**	Preventive Maintenance Inspection
Department	Mechanical Maintenance	**Inspection Date**	28 August 2026
Equipment	Cooling Water Pump P-101A	**Location**	Cooling Water Pump House
Inspection Team	Mechanical Maintenance Team	**Operating Status**	In service during initial inspection

1. PURPOSE OF INSPECTION

The inspection was conducted as part of the preventive maintenance program for Cooling Water Pump P-101A. The objective was to assess the mechanical condition of the pump, motor, coupling, bearing assembly, mechanical seal, and associated piping connections. The inspection also aimed to identify abnormal operating conditions requiring corrective maintenance or further technical evaluation.

2. EQUIPMENT DETAILS

Equipment: Cooling Water Pump P-101A
Pump Type: Centrifugal pump
Service: Cooling water circulation
Rated Flow: 180 m³/h
Rated Head: 52 m
Motor Rating: 45 kW
Motor Speed: 1480 RPM
Pump Driver: Electric motor
Coupling: Flexible coupling
Bearing Arrangement: Rolling-element bearings

The equipment was operating at approximately normal process conditions during the initial inspection.

3. VISUAL INSPECTION

1. Minor leakage was observed around the mechanical seal area.
2. No visible cracks were observed on the pump casing.
3. The pump foundation and anchor bolts were visually inspected and no obvious looseness was observed.
4. Minor surface corrosion was observed around the coupling guard.
5. The flexible coupling guard was found to be installed and secured.
6. No significant external damage was observed on the motor housing.
7. Cable connections at the motor terminal box appeared visually intact.
8. A small amount of oil residue was observed near the bearing housing.

4. VIBRATION MEASUREMENTS

Vibration measurements were recorded at multiple locations.

Measurement Point	Horizontal	Vertical	Axial
Motor DE	3.2 mm/s	3.6 mm/s	2.1 mm/s
Motor NDE	2.8 mm/s	3.1 mm/s	1.9 mm/s
Pump DE	4.6 mm/s	4.8 mm/s	3.2 mm/s
Pump NDE	4.1 mm/s	4.4 mm/s	2.9 mm/s

The highest recorded vibration value was 4.8 mm/s at the Pump DE vertical measurement point. The inspection team recommended further investigation of the vibration condition, particularly at the pump drive-end bearing and coupling area.

5. TEMPERATURE MEASUREMENTS

Bearing temperatures were recorded during operation. The Pump DE bearing showed the highest recorded temperature of 71°C. The inspection team recommended continued monitoring of the bearing temperature and inspection of lubrication

condition during the next maintenance opportunity.

Location	Temperature
Motor DE Bearing	63°C
Motor NDE Bearing	59°C
Pump DE Bearing	71°C
Pump NDE Bearing	68°C

6. MECHANICAL SEAL INSPECTION

The mechanical seal area was inspected for leakage. A minor leakage was observed during operation. No significant leakage was observed from the associated cooling-water piping connections. The condition of the mechanical seal should be monitored. Replacement may be considered if leakage increases or if the seal condition deteriorates. No evidence was recorded indicating a complete mechanical seal failure.

7. COUPLING INSPECTION

The flexible coupling and coupling guard were visually inspected. The coupling guard was secure. However, the inspection team recommended checking coupling alignment during the next scheduled maintenance activity because of the elevated vibration observed at the pump drive-end measurement point. No alignment measurement was recorded during this inspection.

8. BEARING INSPECTION

The pump bearing housing was inspected externally. The Pump DE bearing recorded a temperature of 71°C. A small amount of oil residue was observed near the bearing housing. No internal bearing damage was confirmed during the inspection because the bearing assembly was not dismantled. The inspection team recommended checking the lubrication condition and inspecting the bearing during the next planned maintenance shutdown if the abnormal temperature or vibration persists.

9. MOTOR INSPECTION

The motor housing was visually inspected. No significant external damage was observed. The motor terminal box and cable connections appeared visually intact. Motor vibration measurements were lower than those recorded at the pump drive-end. No electrical test results were included in this inspection report.

10. PIPING AND CONNECTIONS

The pump suction and discharge piping connections were visually inspected. No significant external leakage was observed at the piping connections. The inspection team recommended checking the condition of pipe supports during the next detailed maintenance inspection. No pipe-support load measurements were recorded during the current inspection.

11. KEY FINDINGS

- Minor leakage around the mechanical seal.
- Pump DE vibration reached 4.8 mm/s in the vertical direction.
- Pump DE bearing temperature reached 71°C.
- Minor surface corrosion around the coupling guard.
- Oil residue near the bearing housing.
- No visible cracks on the pump casing.
- No significant external leakage from associated piping connections.
- No electrical test results included.
- No coupling alignment measurement recorded.
- No internal bearing damage confirmed because the bearing was not dismantled.

12. RECOMMENDED ACTIONS

Immediate / Short-Term Monitoring:

1. Monitor Pump DE vibration levels.
2. Monitor Pump DE bearing temperature.
3. Monitor mechanical seal leakage.
4. Monitor the condition of the bearing housing and oil residue.

Planned Maintenance:
5. Check coupling alignment during the next scheduled maintenance activity.
6. Inspect the pump DE bearing and lubrication condition if elevated vibration or temperature persists.
7. Assess the mechanical seal condition and replace the seal if leakage increases or deterioration is confirmed.
8. Inspect coupling guard corrosion during the next maintenance activity.
9. Inspect pump pipe supports during the next detailed maintenance inspection.

13. LIMITATIONS OF INSPECTION

The pump was not dismantled. No internal bearing inspection was performed. No coupling alignment measurement was performed. No electrical test results were obtained. No laboratory analysis of lubricant was performed. No additional vibration spectrum analysis was included in this report. Therefore, internal mechanical conditions that cannot be confirmed through external inspection should not be treated as confirmed findings.

14. PROPOSED MAINTENANCE PRIORITY

Priority 1: Monitor the Pump DE vibration and bearing temperature.
Priority 2: Investigate coupling alignment and bearing/lubrication condition during the next maintenance opportunity.
Priority 3: Monitor mechanical seal leakage and plan replacement if deterioration is observed.
Priority 4: Address minor corrosion and inspect pipe supports during scheduled maintenance.

15. APPROVAL STATUS

This report contains inspection findings and maintenance recommendations.

No maintenance action is recorded as approved in this inspection report.

Any maintenance approval should be issued through the organization's applicable approval process.

16. SOURCE DOCUMENT

Primary Source: IR-CWP-2026-014 — Preventive Maintenance Inspection Report
Equipment: Cooling Water Pump P-101A
Inspection Date: 28 August 2026